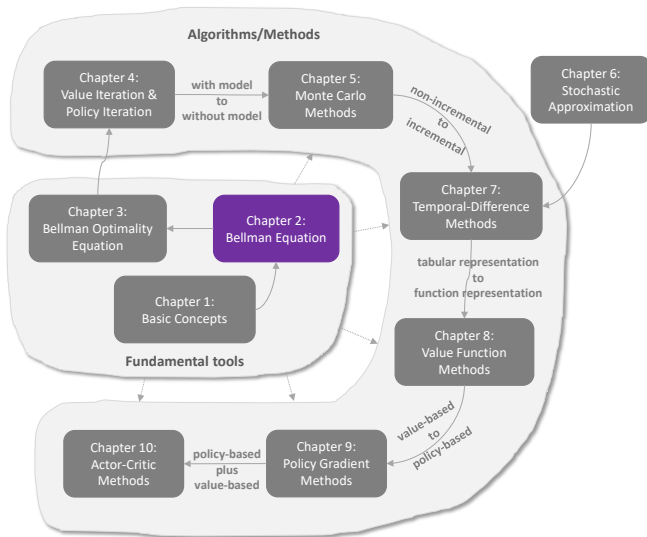


Lecture 2: State Value and Bellman Equation

Shiyu Zhao

Department of Artificial Intelligence
Westlake University

Outline



In this lecture:

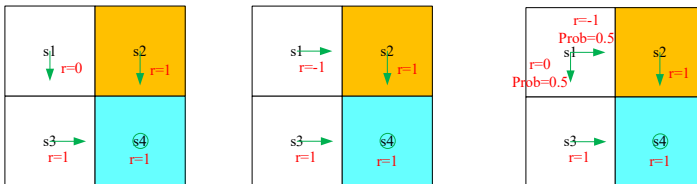
- A core concept: state value
- A fundamental tool: Bellman equation

- 1 Motivating examples
- 2 State value
- 3 Bellman equation: Derivation
- 4 Bellman equation: Matrix-vector form
- 5 Bellman equation: Solve the state values
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Motivating example 1: Why is return important?

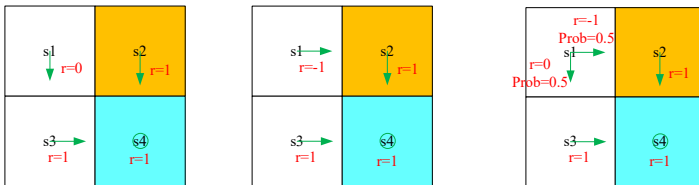
- What is return? The (discounted) sum of the rewards obtained along a trajectory.
- Why is return important? See the following examples.



- Question: Which policy is the “best”? Which is the “worst”?
 - Intuition: the first is the best and the second is the worst, because of the forbidden area.
 - Math: can we use mathematics to describe such intuition?
Return could be used to evaluate policies. See the following.

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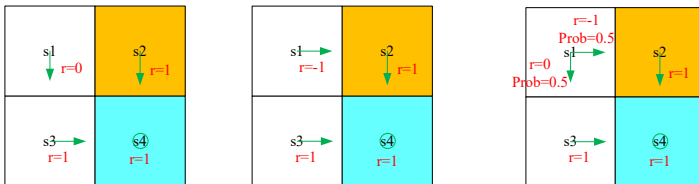
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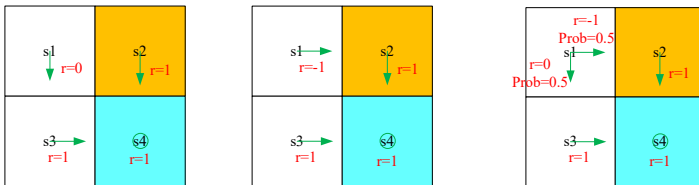
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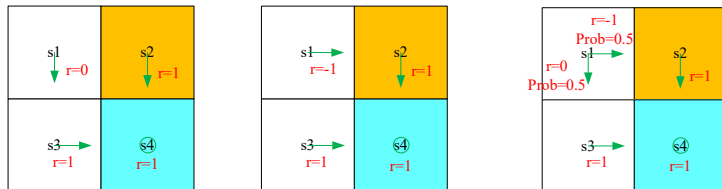
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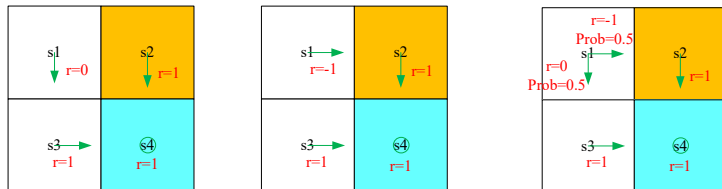
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Based on policy 1 (left figure), starting from s_1 , the discounted return is

$$\begin{aligned}
 \text{return}_1 &= 0 + \gamma 1 + \gamma^2 1 + \dots \\
 &= \gamma(1 + \gamma + \gamma^2 + \dots) \\
 &= \frac{\gamma}{1 - \gamma}
 \end{aligned}$$

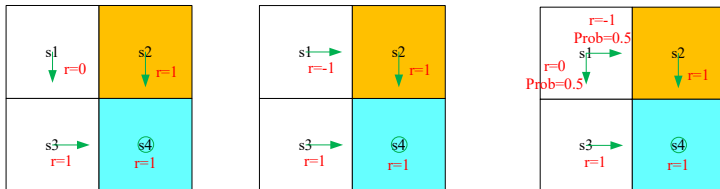
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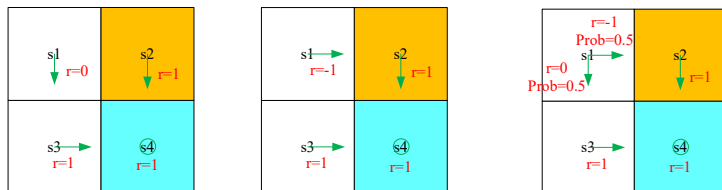


Exercise: Based on policy 2 (middle figure), starting from s_1 , what is the discounted return?

Answer:

$$\begin{aligned}\text{return}_2 &= -1 + \gamma 1 + \gamma^2 1 + \dots \\ &= -1 + \gamma(1 + \gamma + \gamma^2 + \dots) \\ &= -1 + \frac{\gamma}{1 - \gamma}\end{aligned}$$

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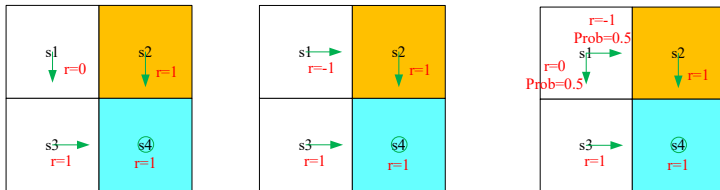


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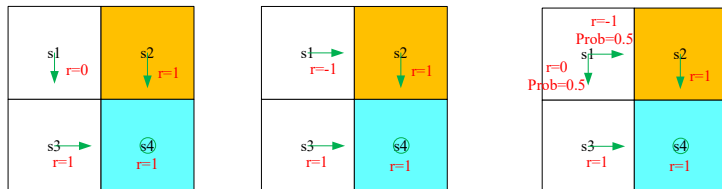
Policy 3 is stochastic!

Exercise: Based on policy 3 (right figure), starting from s_1 , the discounted return is

Answer:

$$\begin{aligned}\text{return}_3 &= 0.5 \left(-1 + \frac{\gamma}{1-\gamma} \right) + 0.5 \left(\frac{\gamma}{1-\gamma} \right) \\ &= -0.5 + \frac{\gamma}{1-\gamma}\end{aligned}$$

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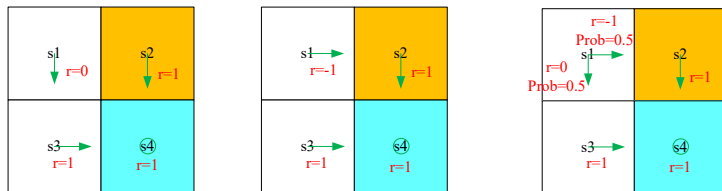
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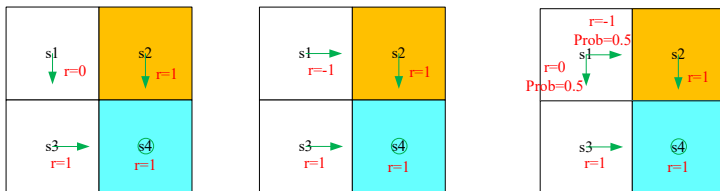


In summary, starting from s_1 ,

$$\text{return}_1 > \text{return}_3 > \text{return}_2$$

- The above inequality suggests that the first policy is the best and the second policy is the worst, which is exactly the same as our intuition.
- Calculating return is important to evaluate a policy.

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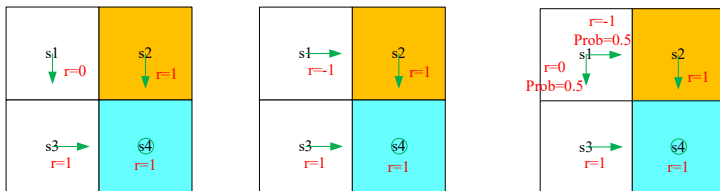


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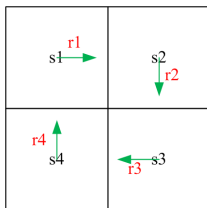
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Motivating example 2: How to calculate return?

While return is important, how to calculate it?



Method 1: by definition

Let v_i denote the return obtained starting from s_i ($i = 1, 2, 3, 4$)

$$v_1 = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots$$

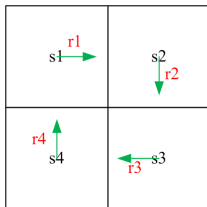
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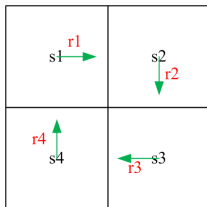
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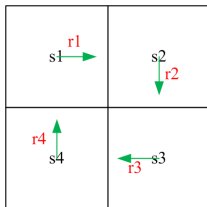
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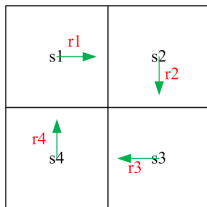
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- The returns rely on each other: Bootstrapping

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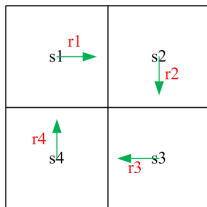
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How to solve these equations? Write in the following **matrix-vector form**:

$$\underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}}_{\mathbf{v}} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix} + \begin{bmatrix} \gamma v_2 \\ \gamma v_3 \\ \gamma v_4 \\ \gamma v_1 \end{bmatrix} = \underbrace{\begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix}}_{\mathbf{r}} + \gamma \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{P}} \underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}}_{\mathbf{v}}$$

which can be rewritten as

$$\mathbf{v} = \mathbf{r} + \gamma \mathbf{P} \mathbf{v}$$

This is the Bellman equation (for this specific deterministic problem)!!

- Though simple, it demonstrates the core idea: the value of one state relies on the values of other states.
- A matrix-vector form is more clear to see how to solve the state values.

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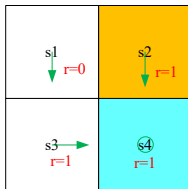
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Exercise: Consider the policy shown in the figure. Please write out the relation among the returns (that is to write out the Bellman equation)



Answer:

$$v_1 = 0 + \gamma v_3$$

$$v_2 = 1 + \gamma v_4$$

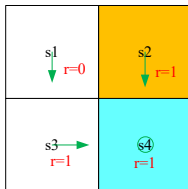
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Consider the following single-step process:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1}$$

- $t, t + 1$: discrete time instances
- S_t : state at time t
- A_t : the action taken in state S_t
- R_{t+1} : the reward obtained after taking A_t
- S_{t+1} : the state transited to after taking A_t

Note that S_t, A_t, R_{t+1} are all *random variables*.

Consider the following **single-step** process:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1}$$

This step is governed by the following probability distributions:

- $S_t \rightarrow A_t$ is governed by $\pi(A_t = a | S_t = s)$
- $S_t, A_t \rightarrow R_{t+1}$ is governed by $p(R_{t+1} = r | S_t = s, A_t = a)$
- $S_t, A_t \rightarrow S_{t+1}$ is governed by $p(S_{t+1} = s' | S_t = s, A_t = a)$

At this moment, we assume we know the model (i.e., the probability distributions)!

Consider the following **multi-step** trajectory:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1} \xrightarrow{A_{t+1}} R_{t+2}, S_{t+2} \xrightarrow{A_{t+2}} R_{t+3}, \dots$$

The discounted return is

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

- $\gamma \in (0, 1)$ is the discount rate.
- G_t is also a random variable since $R_{t+1}, R_{t+2}, \dots$ are random variables.

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The expectation (or called expected value or mean) of G_t is defined as the **state-value function** or simply **state value**:

$$v_{\pi}(s) = \mathbb{E}[G_t | S_t = s]$$

Remarks:

- It is a function of s . It is a conditional expectation with the condition that the state starts from s .
- It is based on the policy π . For a different policy, the state value may be different.

Q: What is the relationship between return and state value?

A: The state value is the mean of all possible returns that can be obtained starting from a state. If everything - $\pi(a|s)$, $p(r|s, a)$, $p(s'|s, a)$ - is deterministic, then state value is the same as return.

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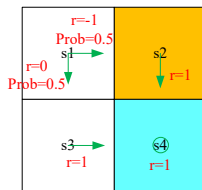
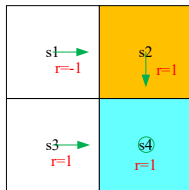
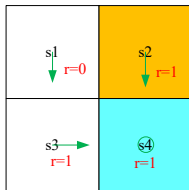
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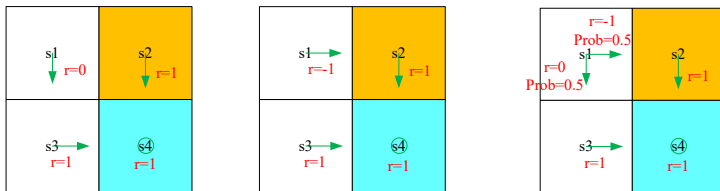
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Example: which policy is good, which is bad?



Recall the returns obtained from s_1 for the three examples:

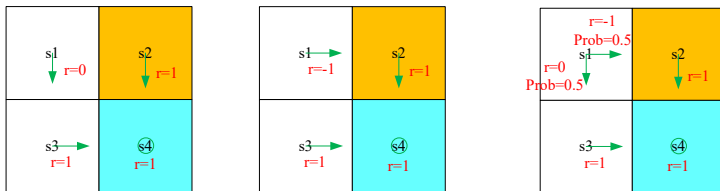
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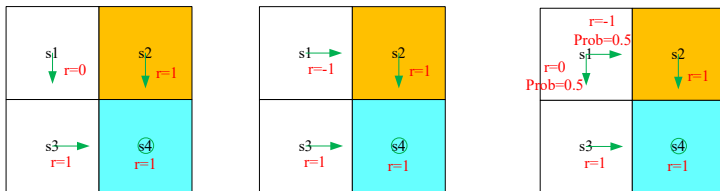


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Example: which policy is good, which is bad?



Recall the returns obtained from s_1 for the three examples:

$$v_{\pi_1}(s_1) = 0 + \gamma 1 + \gamma^2 1 + \dots = \gamma(1 + \gamma + \gamma^2 + \dots) = \frac{\gamma}{1 - \gamma}$$

$$v_{\pi_2}(s_1) = -1 + \gamma 1 + \gamma^2 1 + \dots = -1 + \gamma(1 + \gamma + \gamma^2 + \dots) = -1 + \frac{\gamma}{1 - \gamma}$$

$$v_{\pi_3}(s_1) = 0.5 \left(-1 + \frac{\gamma}{1 - \gamma} \right) + 0.5 \left(\frac{\gamma}{1 - \gamma} \right) = -0.5 + \frac{\gamma}{1 - \gamma}$$

- 1 Motivating examples
- 2 State value
- 3 Bellman equation: Derivation**
- 4 Bellman equation: Matrix-vector form
- 5 Bellman equation: Solve the state values
- 6 Action value
- 7 Summary

- While state value is important, how to calculate? The answer lies in the Bellman equation.
- In a nutshell, the Bellman equation describes the relationship among the values of all states.
- Next, we derive the Bellman equation.
 - There is some math, but don't worry as we already have the intuition.

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Deriving the Bellman equation

Consider a random trajectory:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1} \xrightarrow{A_{t+1}} R_{t+2}, S_{t+2} \xrightarrow{A_{t+2}} R_{t+3}, \dots$$

The return G_t can be written as

$$\begin{aligned} G_t &= R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots, \\ &= R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+3} + \dots), \\ &= R_{t+1} + \gamma G_{t+1} \end{aligned}$$

Then, it follows from the definition of the state value that

$$\begin{aligned} v_\pi(s) &= \mathbb{E}[G_t | S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} | S_t = s] \\ &= \mathbb{E}[R_{t+1} | S_t = s] + \gamma \mathbb{E}[G_{t+1} | S_t = s] \end{aligned}$$

Next, calculate the two terms one by one.

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First, calculate the first term $\mathbb{E}[R_{t+1}|S_t = s]$:

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- This is the mean of *immediate rewards*

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Deriving the Bellman equation

Therefore, we have

$$\begin{aligned} v_{\pi}(s) &= \mathbb{E}[R_{t+1}|S_t = s] + \gamma \mathbb{E}[G_{t+1}|S_t = s], \\ &= \underbrace{\sum_a \pi(a|s) \sum_r p(r|s, a)r}_{\text{mean of immediate rewards}} + \gamma \underbrace{\sum_a \pi(a|s) \sum_{s'} p(s'|s, a)v_{\pi}(s')}_{\text{mean of future rewards}}, \\ &= \sum_a \pi(a|s) \left[\sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v_{\pi}(s') \right], \quad \forall s \in \mathcal{S}. \end{aligned}$$

Highlights:

- The above equation is called the *Bellman equation*, which characterizes the relationship among the state-value functions of different states.
- It consists of two terms: the immediate reward term and the future reward term.
- A set of equations: every state has an equation like this!!!

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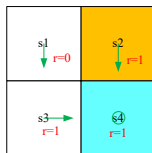
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Highlights: symbols in this equation

- $v_{\pi}(s)$ and $v_{\pi}(s')$ are state values to be calculated. Bootstrapping!
- $\pi(a|s)$ is a given policy. Solving the equation is called policy evaluation.
- $p(r|s, a)$ and $p(s'|s, a)$ represent the dynamic model. What if the model is known or unknown?

An illustrative example



Write out the Bellman equation according to the general expression:

$$v_{\pi}(s) = \sum_a \pi(a|s) \left[\sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v_{\pi}(s') \right]$$

This example is simple because the policy is deterministic.

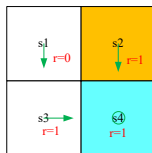
First, consider the state value of s_1 :

- $\pi(a = a_3|s_1) = 1$ and $\pi(a \neq a_3|s_1) = 0$.
- $p(r = 0|s_1, a_3) = 1$ and $p(r \neq 0|s_1, a_3) = 0$.
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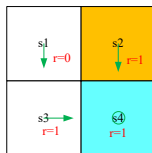
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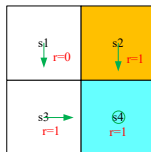
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Similarly, it can be obtained that

$$v_{\pi}(s_1) = 0 + \gamma v_{\pi}(s_2),$$

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An illustrative example

How to solve them?

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Solve the above equations one by one from the last to the first:

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An illustrative example

If $\gamma = 0.9$, then

$$v_{\pi}(s_4) = \frac{1}{1 - 0.9} = 10,$$

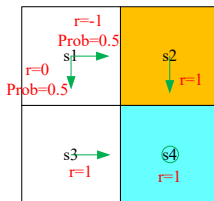
$$v_{\pi}(s_3) = \frac{1}{1 - 0.9} = 10,$$

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$$v_{\pi}(s_1) = \frac{0.9}{1 - 0.9} = 9.$$

What to do after we have calculated state values? Be patient (calculating action value and improve policy)

Exercise



Exercise:

$$v_{\pi}(s) = \sum_a \pi(a|s) \left[\sum_r p(r|s, a) r + \gamma \sum_{s'} p(s'|s, a) v_{\pi}(s') \right]$$

- Write out the Bellman equations for each state.
- Solve the state values from the Bellman equations.
- Compare with the policy in the last example.

Answer:

$$v_{\pi}(s_1) = 0.5[0 + \gamma v_{\pi}(s_3)] + 0.5[-1 + \gamma v_{\pi}(s_2)],$$

$$v_{\pi}(s_2) = 1 + \gamma v_{\pi}(s_4),$$

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Solve the above equations one by one from the last to the first.

$$v_{\pi}(s_4) = \frac{1}{1 - \gamma}, \quad v_{\pi}(s_3) = \frac{1}{1 - \gamma}, \quad v_{\pi}(s_2) = \frac{1}{1 - \gamma},$$

$$\begin{aligned} v_{\pi}(s_1) &= 0.5[0 + \gamma v_{\pi}(s_3)] + 0.5[-1 + \gamma v_{\pi}(s_2)], \\ &= -0.5 + \frac{\gamma}{1 - \gamma}. \end{aligned}$$

Substituting $\gamma = 0.9$ yields

$$v_{\pi}(s_4) = 10, \quad v_{\pi}(s_3) = 10, \quad v_{\pi}(s_2) = 10, \quad v_{\pi}(s_1) = -0.5 + 9 = 8.5.$$

Compare with the previous policy. This one is worse.

- 1 Motivating examples
- 2 State value
- 3 Bellman equation: Derivation
- 4 Bellman equation: Matrix-vector form**
- 5 Bellman equation: Solve the state values
- 6 Action value
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Matrix-vector form of the Bellman equation

Why consider the matrix-vector form? Because we need to solve the state values from it!

- One unknown relies on another unknown. How to solve the unknowns?

$$v_{\pi}(s) = \sum_a \pi(a|s) \left[\sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v_{\pi}(s') \right]$$

- Elementwise form: The above *elementwise form* is valid for every state $s \in \mathcal{S}$. That means there are $|\mathcal{S}|$ equations like this!
- Matrix-vector form: If we put all the equations together, we have a set of linear equations, which can be concisely written in a *matrix-vector form*. The matrix-vector form is very elegant and important.

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Recall that:

$$v_{\pi}(s) = \sum_a \pi(a|s) \left[\sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v_{\pi}(s') \right]$$

Rewrite the Bellman equation as

$$v_{\pi}(s) = r_{\pi}(s) + \gamma \sum_{s'} p_{\pi}(s'|s)v_{\pi}(s')$$

where

$$r_{\pi}(s) \doteq \sum_a \pi(a|s) \sum_r p(r|s, a)r, \quad p_{\pi}(s'|s) \doteq \sum_a \pi(a|s)p(s'|s, a)$$

Matrix-vector form of the Bellman equation

Suppose the states could be indexed as s_i ($i = 1, \dots, n$).

For state s_i , the Bellman equation is

$$v_{\pi}(s_i) = r_{\pi}(s_i) + \gamma \sum_{s_j} p_{\pi}(s_j | s_i) v_{\pi}(s_j)$$

Put all these equations for all the states together and rewrite to a matrix-vector form

$$v_{\pi} = r_{\pi} + \gamma P_{\pi} v_{\pi}$$

where

- $v_{\pi} = [v_{\pi}(s_1), \dots, v_{\pi}(s_n)]^T \in \mathbb{R}^n$
- $r_{\pi} = [r_{\pi}(s_1), \dots, r_{\pi}(s_n)]^T \in \mathbb{R}^n$
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Illustrative examples

If there are four states, $v_\pi = r_\pi + \gamma P_\pi v_\pi$ can be written out as

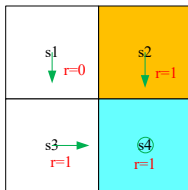
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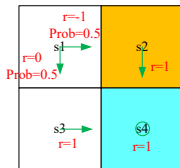
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Illustrative examples

If there are four states, $v_\pi = r_\pi + \gamma P_\pi v_\pi$ can be written out as

$$\underbrace{\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}}_{v_\pi} = \underbrace{\begin{bmatrix} r_\pi(s_1) \\ r_\pi(s_2) \\ r_\pi(s_3) \\ r_\pi(s_4) \end{bmatrix}}_{r_\pi} + \gamma \underbrace{\begin{bmatrix} p_\pi(s_1|s_1) & p_\pi(s_2|s_1) & p_\pi(s_3|s_1) & p_\pi(s_4|s_1) \\ p_\pi(s_1|s_2) & p_\pi(s_2|s_2) & p_\pi(s_3|s_2) & p_\pi(s_4|s_2) \\ p_\pi(s_1|s_3) & p_\pi(s_2|s_3) & p_\pi(s_3|s_3) & p_\pi(s_4|s_3) \\ p_\pi(s_1|s_4) & p_\pi(s_2|s_4) & p_\pi(s_3|s_4) & p_\pi(s_4|s_4) \end{bmatrix}}_{P_\pi} \underbrace{\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}}_{v_\pi}.$$

For the following example:



$$\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix} = \begin{bmatrix} 0.5(0) + 0.5(-1) \\ 1 \\ 1 \\ 1 \end{bmatrix} + \gamma \begin{bmatrix} 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}.$$

- 1 Motivating examples
- 2 State value
- 3 Bellman equation: Derivation
- 4 Bellman equation: Matrix-vector form
- 5 Bellman equation: Solve the state values**
- 6 Action value
- 7 Summary

Why to solve state values?

- Given a policy, finding out the corresponding state values is called **policy evaluation**!
- It is a fundamental problem in RL. It is the foundation to find better policies.
- Therefore, it is important to understand how to solve the Bellman equation.

The Bellman equation in matrix-vector form is

$$v_\pi = r_\pi + \gamma P_\pi v_\pi$$

- The closed-form solution is

$$v_\pi = (I - \gamma P_\pi)^{-1} r_\pi$$

- The matrix $I - \gamma P_\pi$ is invertible. See details in my book.
- We still need to use numerical algorithms to calculate the matrix inverse.
- Can we avoid the matrix inverse operation? Yes, as shown below.

- An iterative solution is

$$v_{k+1} = r_\pi + \gamma P_\pi v_k$$

This algorithm leads to a sequence $\{v_0, v_1, v_2, \dots\}$. We can show that

$$v_k \rightarrow v_\pi = (I - \gamma P_\pi)^{-1} r_\pi, \quad k \rightarrow \infty$$

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Solve state values (optional)

Proof.

Define the error as

$$\delta_k = v_k - v_\pi.$$

We only need to show $\delta_k \rightarrow 0$. Substituting $v_{k+1} = \delta_{k+1} + v_\pi$ and $v_k = \delta_k + v_\pi$ into $v_{k+1} = r_\pi + \gamma P_\pi v_k$ gives

$$\delta_{k+1} + v_\pi = r_\pi + \gamma P_\pi (\delta_k + v_\pi),$$

which can be rewritten as

$$\delta_{k+1} = -v_\pi + r_\pi + \gamma P_\pi \delta_k + \gamma P_\pi v_\pi = \gamma P_\pi \delta_k.$$

As a result,

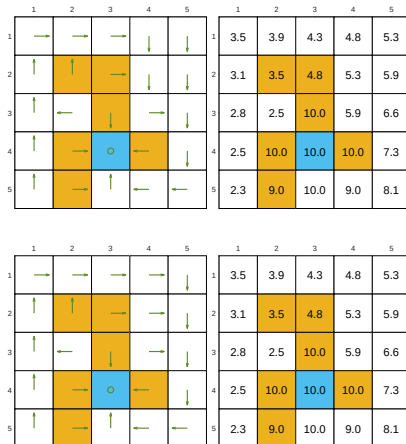
$$\delta_{k+1} = \gamma P_\pi \delta_k = \gamma^2 P_\pi^2 \delta_{k-1} = \dots = \gamma^{k+1} P_\pi^{k+1} \delta_0.$$

Note that $0 \leq P_\pi^k \leq 1$, which means every entry of P_π^k is no greater than 1 for any $k = 0, 1, 2, \dots$. That is because $P_\pi^k \mathbf{1} = \mathbf{1}$, where $\mathbf{1} = [1, \dots, 1]^T$. On the other hand, since $\gamma < 1$, we know $\gamma^k \rightarrow 0$ and hence $\delta_{k+1} = \gamma^{k+1} P_\pi^{k+1} \delta_0 \rightarrow 0$ as $k \rightarrow \infty$. $\square$

Solve state values

Examples: $r_{\text{boundary}} = r_{\text{forbidden}} = -1$, $r_{\text{target}} = +1$, $\gamma = 0.9$

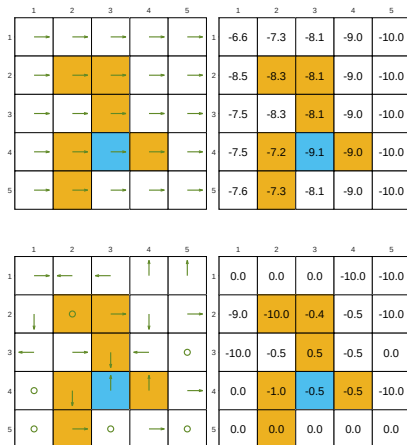
- The following are two “good” policies and the state values. The two policies are different for the top two states in the forth column.



Solve state values

Examples: $r_{\text{boundary}} = r_{\text{forbidden}} = -1$, $r_{\text{target}} = +1$, $\gamma = 0.9$

- The following are two “bad” policies and the state values. The state values are less than those of the good policies.



- 1 Motivating examples
- 2 State value
- 3 Bellman equation: Derivation
- 4 Bellman equation: Matrix-vector form
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From state value to action value:

- State value: the average return the agent can get *starting from a state*.
- Action value: the average return the agent can get *starting from a state and taking an action*.

Why do we care action value? Because we want to know which action is better. This point will be clearer in the following lectures. We will frequently use action values.

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Definition:

$$q_{\pi}(s, a) = \mathbb{E}[G_t | S_t = s, A_t = a]$$

- $q_{\pi}(s, a)$ is a function of the state-action pair (s, a)
- $q_{\pi}(s, a)$ depends on π

It follows from the properties of conditional expectation that

$$\underbrace{\mathbb{E}[G_t | S_t = s]}_{v_{\pi}(s)} = \sum_a \underbrace{\mathbb{E}[G_t | S_t = s, A_t = a]}_{q_{\pi}(s, a)} \pi(a|s)$$

Hence,

$$v_{\pi}(s) = \sum_a \pi(a|s) q_{\pi}(s, a) \tag{1}$$

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Recall that the state value is given by

$$v_{\pi}(s) = \sum_a \pi(a|s) \left[\underbrace{\sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v_{\pi}(s')}_{q_{\pi}(s, a)} \right] \quad (2)$$

By comparing (1) and (2), we have the **action-value function** as

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(1) and (3) are the two sides of the same coin:

- (1) shows how to obtain state values from action values.
- (3) shows how to obtain action values from state values.

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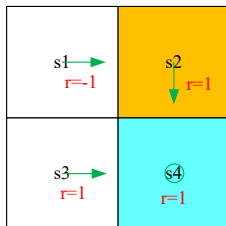
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Illustrative example for action value



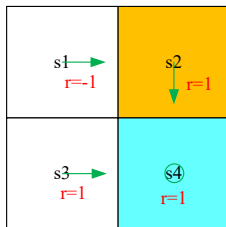
Write out the action values for state s_1 :

$$q_{\pi}(s_1, a_2) = -1 + \gamma v_{\pi}(s_2),$$

Questions:

- $q_{\pi}(s_1, a_1), q_{\pi}(s_1, a_3), q_{\pi}(s_1, a_4), q_{\pi}(s_1, a_5) = ?$ Be careful!

Illustrative example for action value



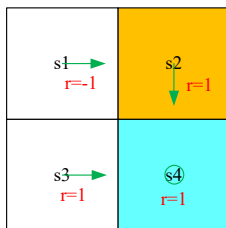
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Illustrative example for action value



For the other actions:

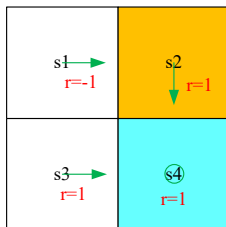
$$q_{\pi}(s_1, a_1) = -1 + \gamma v_{\pi}(s_1),$$

$$q_{\pi}(s_1, a_3) = 0 + \gamma v_{\pi}(s_3),$$

$$q_{\pi}(s_1, a_4) = -1 + \gamma v_{\pi}(s_1),$$

$$q_{\pi}(s_1, a_5) = 0 + \gamma v_{\pi}(s_1).$$

Illustrative example for action value



Highlights:

- Action value is important since we care about which action to take.
- We can first calculate all the state values and then calculate the action values.
- We can also directly calculate the action values with or without models.

- 1 Motivating examples
- 2 State value
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Key concepts and results:

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- Action value: $q_\pi(s, a) = \mathbb{E}[G_t | S_t = s, A_t = a]$
- The Bellman equation (elementwise form):

$$\begin{aligned} v_\pi(s) &= \sum_a \pi(a|s) \left[\underbrace{\sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v_\pi(s')}_{q_\pi(s, a)} \right] \\ &= \sum_a \pi(a|s) q_\pi(s, a) \end{aligned}$$

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- How to solve the Bellman equation: closed-form solution, iterative solution

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